

## Author

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## 1 Introduction

The context to be considered is movies.

| idimdb    | title                           | release    | runtime | budget  | revenue   | poster                                                | overview                                              | rating | last_update |
|-----------|---------------------------------|------------|---------|---------|-----------|-------------------------------------------------------|-------------------------------------------------------|--------|-------------|
| tt0029583 | Snow White and the Seven Dwarfs | 1937-12-21 | 83      | 1488423 | 184925486 | https://image.tmdb.org/t/p/original/1P9eGGIT7eV7kA... | A beautiful girl, Snow White, takes refuge in the ... | 7.6    | 2022-02-27  |
| tt0032910 | Pinocchio                       | 1940-02-23 | 88      | 2600000 | 84300000  | https://image.tmdb.org/t/p/original/ynkEEi296ofs9k... | Lonely toymaker Geppetto has his wishes answered w... | 7.4    | 2022-02-27  |
| tt0033563 | Dumbo                           | 1941-10-31 | 64      | 812000  | 1600000   | https://image.tmdb.org/t/p/original/hKDdlisIMtsU9J... | Dumbo is a baby elephant born with over-sized ears... | 7.2    | 2022-02-27  |
| tt0034492 | Bambi                           | 1942-08-14 | 70      | 858000  | 267447150 | https://image.tmdb.org/t/p/original/wV9e2y4myJ4KMF... | Bambi's tale unfolds from season to season as the ... | 7.3    | 2022-02-27  |
| tt0042332 | Cinderella                      | 1950-02-22 | 74      | 2900000 | 263591415 | https://image.tmdb.org/t/p/original/4nssBcQUBadCTB... | Cinderella has faith her dreams of a better life w... | 7.3    | 2022-02-27  |
| tt0046183 | Peter Pan                       | 1953-02-05 | 77      | 4000000 | 87404651  | https://image.tmdb.org/t/p/original/fJJOs1yrtKIZc...  | Leaving the safety of their nursery behind, Wendy,... | 7.3    | 2022-07-13  |
| tt0048280 | Lady and the Tramp              | 1955-06-22 | 76      | 4000000 | 36359037  | https://image.tmdb.org/t/p/original/wXKeGee4htM7Yz... | Lady, a golden cocker spaniel, meets up with a mon... | 7.3    | 2022-10-18  |
| tt0053285 | Sleeping Beauty                 | 1959-02-17 | 75      | 6000000 | 51600000  | https://image.tmdb.org/t/p/original/j2mTyUukcLwDle... | A beautiful princess born in a faraway kingdom is ... | 7.2    | 2022-02-27  |
| tt0055254 | One Hundred and One Dalmatians  | 1961-01-25 | 79      | 4000000 | 85000000  | https://image.tmdb.org/t/p/original/9kiDisS1sVb5LJ... | When a litter of dalmatian puppies are abducted by... | 7.3    | 2022-02-27  |
| tt0057546 | The Sword in the Stone          | 1963-12-25 | 79      | 3000000 | 22182353  | https://image.tmdb.org/t/p/original/7lyeeuhGAJSNX...  | Wart is a young boy who aspires to be a knight's s... | 7.2    | 2022-02-27  |
| tt0058331 | Mary Poppins                    | 1964-08-27 | 139     | 6000000 | 103082380 | https://image.tmdb.org/t/p/original/yO8UyO86uNBq12... | A magical nanny employs music and adventure to hel... | 7.8    | 2022-02-27  |

Figura 1: Sample

## 2 Deployment

Docker pull command

```
docker pull elasticsearch:8.17.10
```

```
docker pull elasticsearch:7.17.10
```

Server (first time)

```
docker run --name my-es \
  -p 9200:9200 \
  -p 9300:9300 \
  -e "discovery.type=single-node" \
  -e "xpack.security.enabled=false" \
  -e "ES_JAVA_OPTS=-Xms1g -Xmx1g" \
  elasticsearch:8.17.10
```

```
docker run -p 9200:9200 -p 9300:9300 -e "discovery.type=single-node" --name my-es
  elasticsearch:7.17.10
```

Server (if already exists)

```
docker start my-es
```

Checking server is alive on <http://localhost:9200>

## 3 Tutorial

### 3.1 Index creation

Download the following files:

1. movies.csv [https://people.irisa.fr/Laurent.Dorazio/data/teachings/big\\_data/es/movies.csv](https://people.irisa.fr/Laurent.Dorazio/data/teachings/big_data/es/movies.csv)
2. movies.sh [https://people.irisa.fr/Laurent.Dorazio/data/teachings/big\\_data/es/movies.sh](https://people.irisa.fr/Laurent.Dorazio/data/teachings/big_data/es/movies.sh)
3. csv2ndjson.py [https://people.irisa.fr/Laurent.Dorazio/data/teachings/big\\_data/es/csv2ndjson.py](https://people.irisa.fr/Laurent.Dorazio/data/teachings/big_data/es/csv2ndjson.py)

The first one consists in a set of movies as a CSV files. The second one is a script enabling to build and then query an index with these movies. In particular the script includes a schema definition:

---

```
"mappings": {
  "properties":{
    "idimdb": { "type": "text" },
    "title": { "type": "text", "fields": { "keyword": { "type": "keyword" } } },
    "release": { "type": "date" },
    "runtime": { "type": "integer" },
    "budget": { "type": "integer" },
    "revenue": { "type": "integer" },
    "poster": { "type": "text" },
    "rating": { "type": "float" }
  }
}
```

---

It has to be noted that the script csv2ndjson.py is required by movies.sh to convert the data from the proposed CSV format to the NDJSON (Newline delimited JSON) format used to insert data in the ElasticSearch server. Executing the following command (in a bash shell) enables to execute the script.

---

```
source ./movies.sh
```

---

The scripts provides example of URL and DSL queries.

Example of URL query (with *user1* being the current user)

[http://localhost:9200/user1-movies/\\_search?q=runtime:60&pretty](http://localhost:9200/user1-movies/_search?q=runtime:60&pretty)

Example DSL query

---

```
function ex_dsl_0() {
  titre "Less than 70 minutes movies"
  curljson -X GET "${INDEX}/_search?pretty" <<JSON
  {
    "query": {
      "range": {
        "runtime": { "lt": 70}
      }
    }
  }
}
JSON
}
```

---

Note that you have to comment/uncomment the relevant instructions.

## 4 Activities

Write the following queries:

### 4.1 URL

1. 82 minutes movies
2. Movies which title contains christmas
3. Movies which title contains christmas or prince
4. Movies which title contains christmas and prince
5. Movies which budget greater than 250000000

6. Movies which runtime is between 65 and 75 (2 versions)
7. Movies which title contains prince, limited to 2 results
8. Movies which title contains star but not wars
9. Movies which title is star wars
10. One 90-minute movie
11. 95-minute movies which title does not contain 2 and a budget greater than or equal to \$150000000
12. Shortest movie

## 4.2 DSL

1. Longer than 170 minutes movies
2. Movies which title contains star
3. Movies which title is Aladdin
4. Movies which title contains star or wars
5. Movies which title contains star and not wars
6. Movies which title contains star and wars (both of them)
7. 136 minute movies which title contains star
8. 136 or 60 minute movies
9. Average runtime
10. Budget of the most expensive movie
11. Movies sorted per revenue
12. Distinct runtimes

## 4.3 More queries

1. Movie with a runtime equal to 90 or 120 and a budget different than 28000000 which title contains Puss or Panda but not Romeo (2 versions in URL and at least one with + or -)
2. Average runtime of the movie with a budget greater than 1000000.

## 5 References

- **Lecture:** [https://people.irisa.fr/Laurent.Dorazio/data/teachings/big\\_data/big\\_data\\_es.pdf](https://people.irisa.fr/Laurent.Dorazio/data/teachings/big_data/big_data_es.pdf)
- **ElasticSearch:** <https://www.elastic.co/fr/>
- **Pierre Nerzic class on Big Data:** <https://perso.univ-rennes1.fr/pierre.nerzic/Hadoop/>